

EE236: Sub-threshold Characteristics of N-channel MOSFET

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1 Aim of the experiment

- Obtain the sub-threshold I_d vs V_{gs} characteristics of NMOS.
- Check the continuity of I_d vs V_{gs} characteristics beyond the sub-threshold region.
- Employing the NMOS to be used as a Common Source Amplifier in sub-threshold region.

2 Theory

An N-channel MOSFET is a Field Effect Transistor that has 4 terminals: 1) Drain, 2) Gate, 3) Source, and 4) Body. The body terminal of an NMOS is connected to the lowest voltage possible in the circuit, typically ground. For an NMOS to be "ON," the applied Gate-to-Source Voltage must be greater than the Threshold Voltage V_{th} ; otherwise, it is assumed to be "OFF," i.e., $V_{gs} \geq V_{th}$.

In reality, however, the MOSFET is not completely "OFF" but conducts a small amount of current. When the Gate-to-Source Voltage is less than the threshold voltage, a few electrons are still present inside the channel. When a Drain Voltage is applied, a depletion region forms near the drain junction, and there is a lower concentration of minority carriers.

As a result, electrons near the Source Junction start moving towards the Drain Junction, primarily due to diffusion, leading to Subthreshold Current. The equation for the Drain current in the Subthreshold region is given by:

$$I_d = I_0 e^{\frac{(V_{gs}-V_{th})}{\eta V_t}} \left(1 - e^{\frac{-V_{ds}}{\eta V_t}}\right) \quad (1)$$

where

$$I_0 = \mu_n C_{ox} \frac{W}{L} V_t^2 (\eta - 1) \quad (2)$$

Here: - I_d is the Drain Current, - V_{gs} is the Gate-to-Source Voltage, - V_{th} is the Threshold Voltage of the NMOS, - V_t is the Thermal Voltage, - V_{ds} is the Drain-to-Source Voltage, - μ_n is the Mobility of electrons, - C_{ox} is the Gate Oxide Capacitance per unit area, - $\frac{W}{L}$ is the Aspect Ratio of the NMOS, - η is the Subthreshold Slope Factor, given by

$$\eta = 1 + \frac{C_d}{C_{ox}} \quad (3)$$

where C_d is the Depletion Capacitance.

We start with an assumed value of η , say $\eta = 10$. At room temperature, the value of $\eta V_t = 0.256$. If $V_{ds} = 2$ V, the exponential term containing V_{ds} diminishes to zero. Thus, the overall term containing V_{ds} becomes 1, simplifying the expression. This allows us to approximate the Drain current as:

$$I_d \approx I_0 e^{\frac{(V_{gs}-V_{th})}{\eta V_t}} \quad (4)$$

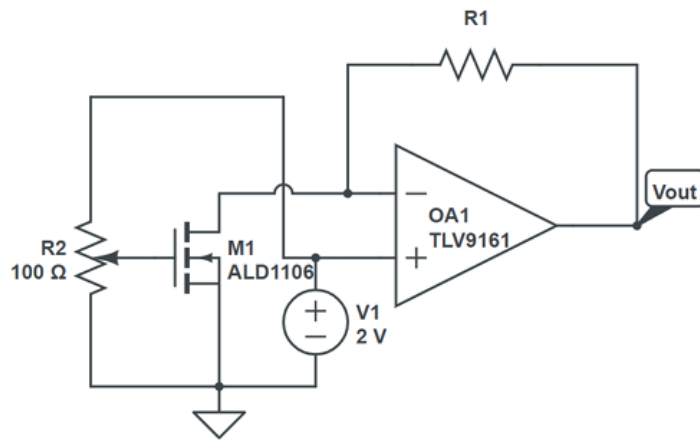
3 Components Necessary

- ALD1106 NMOS I.C.
- TLV9161 OP-AMP
- Keithley Power Supply
- 10 M Ω and 1 M Ω resistor (2 quantities each)
- 100 Ω Potentiometer
- 3 Digital Multimeters (D.M.M.)

4 Experiment

4.1 Part - 1

4.1.1 Circuit Diagram



4.1.2 Results

Vgs (V)	Vout (V)	Id (nA)
0.15	1.88	-12
0.25	1.89	-11
0.3	1.9	-10
0.32	1.91	-9
0.34	1.92	-8
0.36	1.94	-6
0.38	1.98	-2
0.4	2.04	4
0.42	2.13	13
0.44	2.26	26
0.46	2.47	47
0.48	2.77	77
0.5	3.11	111
0.52	3.46	146
0.54	3.5	220

Table 1: Readings for $R_1 = 10\ M\Omega$

Vgs (V)	Vout (V)	Id (nA)
0.56	3.51	370
0.58	3.52	500
0.6	3.52	620
0.62	3.53	720
0.64	3.53	580
0.66	3.54	500
0.68	3.54	570

Table 2: Readings for $R_1 = 1\ M\Omega$

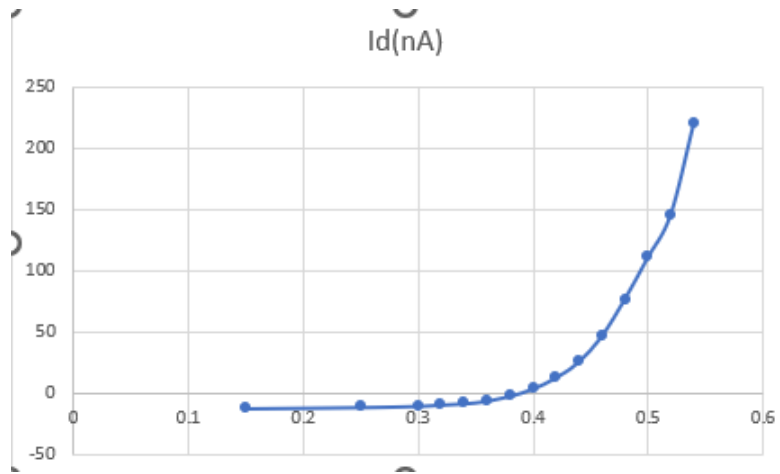


Figure 1: I_d vs V_{gs} for $R_1 = 10\text{ M}\Omega$

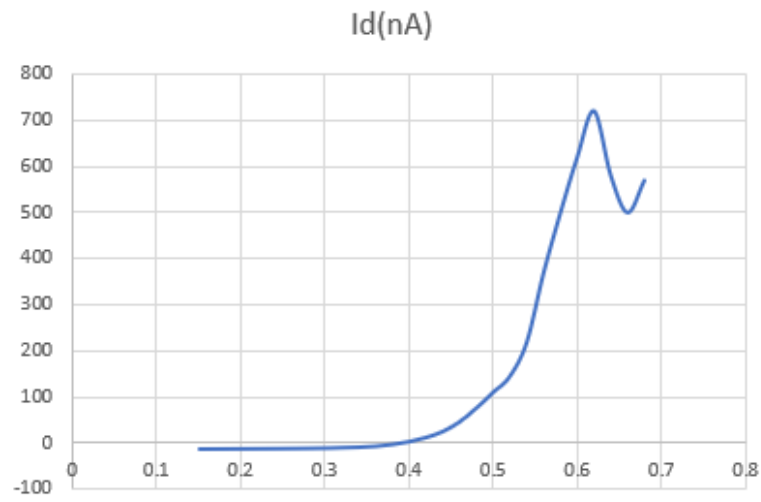


Figure 2: I_d vs V_{gs} (combined data)

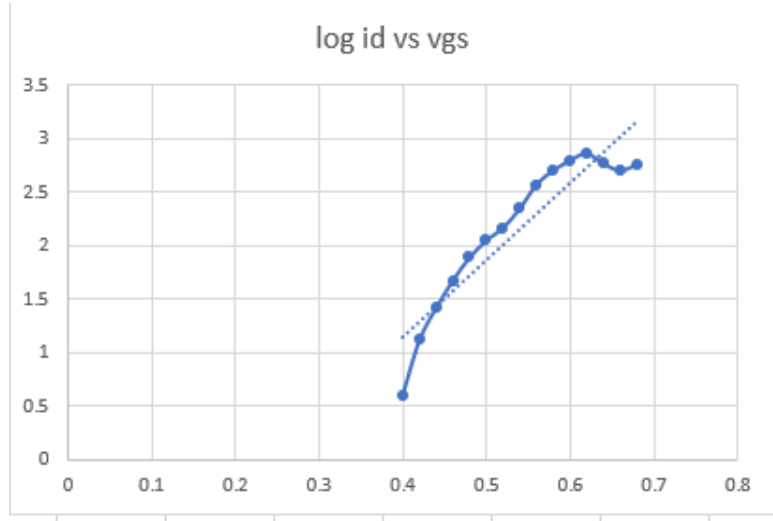


Figure 3: $\ln(I_d)$ vs V_{gs}

4.1.3 Sub Threshold Swing, S

The Sub Threshold Swing, S , is defined as:

$$S = \frac{dV_{gs}}{d(\log_{10} I_d)} \quad (5)$$

The value of S is obtained by taking the slope of the graph $\log_{10} I_d$ vs. V_{gs} and taking the inverse of it. From our plot, $S = 0.307$ mV/decade

For a FET such as the ALD1106, the value of S is given by:

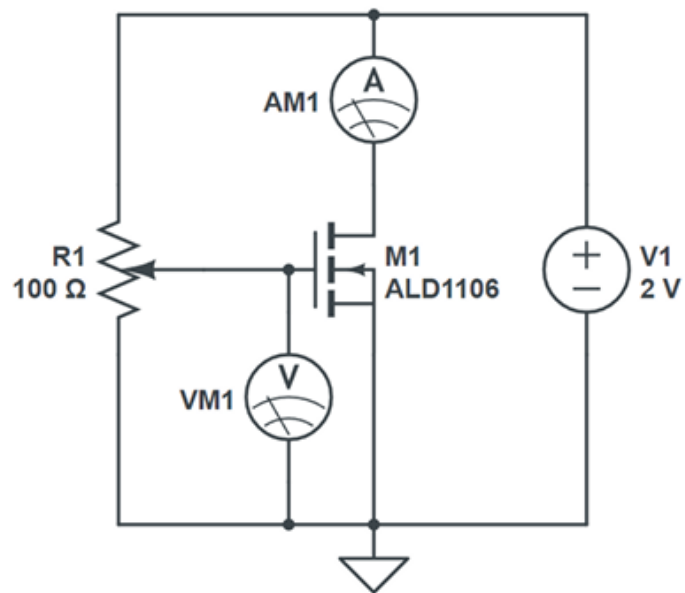
$$S = 60\eta \text{ mV/decade} \quad (6)$$

Therefore. $\eta = 0.005$

$I_{off} = 15nA$ and correspondingly, the power dissipation under $V_{gs} = 0$ is $30nW$

4.2 Part 2 : Graph Continuity

4.2.1 Circuit Diagram



4.2.2 Results

Vgs (V)	Id (nA)
0.7	720
0.72	910
0.74	1120
0.76	1350
0.78	1620
0.8	1900
0.82	2220
0.84	2570
0.86	2950
0.88	3360
0.9	3800
1	6420
1.1	9760
1.2	13810
1.3	18560
1.4	23900
1.5	30100
1.6	36800
1.7	44100
1.8	52100
1.9	60600
2	69600

Table 3: I_d vs V_{gs}

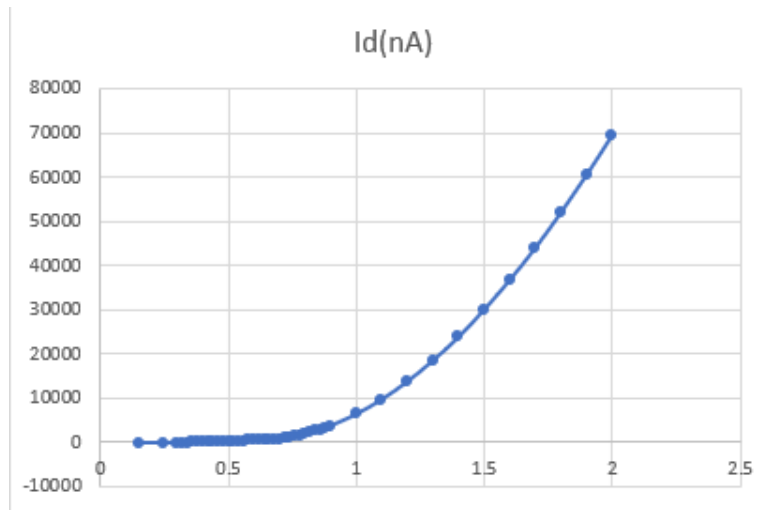
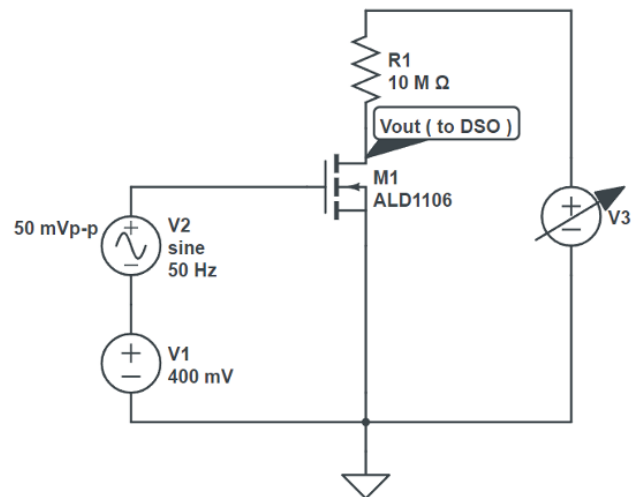


Figure 4: I_d vs V_{gs}

4.3 Part-3 : Common Source Amplifier using MOS FET biased in Sub-Threshold

4.3.1 Circuit Diagram



4.4 Results

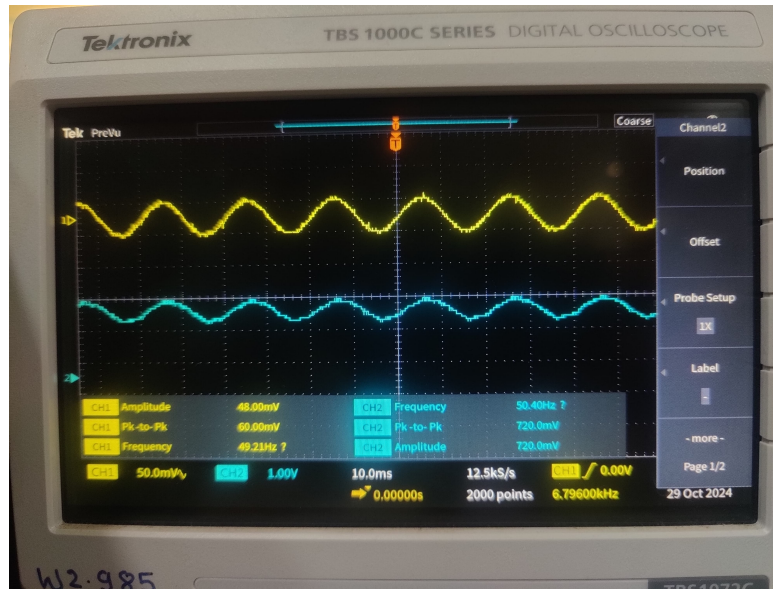


Figure 5: Output waveform vs Input waveform

$$\text{Gain} = \frac{V_{outpp}}{V_{inpp}} = \frac{720mV}{60mV} = 12$$